

[Dashboard](#) / [My courses](#) / [MA-224-G 24H](#) / [Tests](#)/ [Test 3 \(topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of integers mod\(n\) and Lagrange's theorem\)](#)**Started on** Wednesday, 13 November 2024, 1:46 PM**State** Finished**Completed on** Wednesday, 13 November 2024, 1:59 PM**Time taken** 13 mins 42 secs**Marks** 1.00/3.00**Grade** 1.00 out of 3.00 (33.36%)

Information

## Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.



## Question 1

Incorrect

Mark 0.00 out of 1.00

The following sequence

$$p_n = -\frac{(-2)^n}{3}$$

is a special solution to the recurrence relation

$$-5 \cdot a_n - 4 \cdot a_{n-1} = (-2)^n.$$

**a)** Find another solution to the recurrence relation, not equal to  $p_n$ :

$$a_n = -((-2)^2/(3)) + ((-4)/(5))^{(n)}$$

Your last answer was interpreted as follows:

$$-\frac{(-2)^2}{3} + \left(\frac{-4}{5}\right)^n$$

The variables found in your answer were:  $[n]$

**Syntax guide:** Your answer should be an expression which depends on the variable  $n$ , and *no other constants*. E.g. if you think the answer is

$$-7 \cdot \frac{2^n + 1}{3n},$$

you should type **-7\*(2^n+1) / (3\*n)**.

**b)** Given the initial condition  $a_0 = \frac{2}{3}$ . Compute  $a_n$  for  $n = 1$ :

$$a_1 = 0$$

Your last answer was interpreted as follows:

$$0$$



## Question 2

Incorrect

Mark 0.00 out of 1.00

In this problem you are tasked with computing subgroups of  $(\mathbb{Z}/n, +_n)$  and direct products of such groups.

a) Find the subgroup generated by the element  $1 \in \mathbb{Z}/3$ .

$\langle 1 \rangle =$

Your last answer was interpreted as follows:

$$\{[0, 0], [1, 0], [2, 0]\}$$

**Syntax guide:** Input your answer as a set. E.g. if you think the answer is  $\{1, 2, 3\}$ , then you should type **{1,2,3}**.

b) Find the subgroup generated by the element  $2 \in \mathbb{Z}/4$ .

$\langle 2 \rangle =$

Your last answer was interpreted as follows:

$$\{[0, 0], [0, 2]\}$$

**Syntax guide:** Input your answer as a set. E.g. if you think the answer is  $\{1, 2, 3\}$ , then you should type **{1,2,3}**.

c) Find the subgroup generated by the element  $[1, 2] \in \mathbb{Z}/3 \times \mathbb{Z}/4$ .

$\langle [1, 2] \rangle =$

Your last answer was interpreted as follows:

$$\{[0, 0], [1, 2], [2, 0], [0, 2]\}$$

**Syntax guide:** Elements of this group are on the form  $[a, b]$ . Your answer should be a set of such elements. E.g. if you think the answer is  $\{[1, 2], [0, 2], [1, 3]\}$ , then you should type **{[1,2],[0,2],[1,3]}**.



## Question 3

Correct

Mark 1.00 out of 1.00

a) Let  $G = (\mathbb{Z}/2)^{\times 5}$ , and let  $H \subset G$  be the subgroup

$$H = \{[0, 0, 0, 0, 0], [0, 1, 1, 0, 0], [1, 0, 0, 1, 0], [1, 1, 1, 1, 0]\}.$$

What is the index of  $H \subset G$ ?

$$[G : H] = \boxed{8}$$

Your last answer was interpreted as follows:

8

b) Let  $D_4$  be the dihedral group of order 8 of symmetries of the unit square centered at the origin of  $\mathbb{R}^2$ . Let  $\sigma \in D_4$  be the symmetry that reflects around the line  $(x=0)$ .

Compute the index of the subgroup generate by  $\sigma$ :

$$[D_4 : \langle \sigma \rangle] = \boxed{4}$$

Your last answer was interpreted as follows:

4

## Question 4

Correct

Mark 0.00 out of 0.00

## Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 174

Signing code:

Your last answer was interpreted as follows:

18593

[◀ Test 2 \(topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations\)](#)

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